type "%USERPROFILE%\.jenkins\secrets\initialAdminPassword"

PRACTICAL 08

🚀 Step 1: Start Jenkins (Windows)

👉 Open Command Prompt:

```
java -jar jenkins.war
```

👉 Open browser:

```
http://localhost:8080
```

👉 Unlock Jenkins (use password from CMD)

🔌 Step 2: Install Required Plugins

👉 In Jenkins UI:

- Go to **Manage Jenkins → Plugins**
- Install:
- Git Plugin
- GitHub Plugin
- Pipeline Plugin

🌐 Step 3: Create GitHub Repository

👉 Go to GitHub
👉 Create repo: `jenkins-demo`

💻 Step 4: Create Project Files (Windows CMD / Git Bash)

```
mkdir jenkins-demo
cd jenkins-demo
```

👉 Create app.py:

```
notepad app.py
```

```
print("Hello from Jenkins Pipeline")
```

☞ Create requirements.txt:

```
notepad requirements.txt
# empty or add dependencies if needed
```

📄 Step 5: Create Jenkinsfile (VERY IMPORTANT)

```
notepad Jenkinsfile
```

☞ Paste this:

```
pipeline {
    agent any

    stages {
        stage('Build') {
            steps {
                echo 'Building project...'
            }
        }
        stage('Run') {
            steps {
                bat 'python app.py'
            }
        }
    }
}
```

💾 Step 6: Push Code to GitHub

```
git init
git add .
git commit -m "Initial commit for Jenkins"
git branch -M main
```

```
git remote add origin https://github.com/your-username/jenkins-demo.git
git push -u origin main
```

⚙️ Step 7: Create Pipeline Job in Jenkins

👉 In Jenkins UI:

1. Click **New Item**
2. Enter name: `jenkins-job`
3. Select **Pipeline**
4. Click OK

🔗 Step 8: Configure Pipeline

👉 In job settings:

- Go to **Pipeline section**
- Select: **Pipeline script from SCM**
- SCM: **Git**
- Paste repo URL:
  `https://github.com/your-username/jenkins-demo.git`
- Branch: `main`

👉 Click **Save**

🔄 Step 9: Build Trigger

👉 Option 1 (Simple for exam):

- Click **Build Now**

👉 Option 2:

- Enable **Poll SCM**

▶ Step 10: Run Build

☞ Click **Build Now**

☞ Check console output:

- Go to build → Console Output
- You will see:

```
Building project...
Hello from Jenkins Pipeline
```